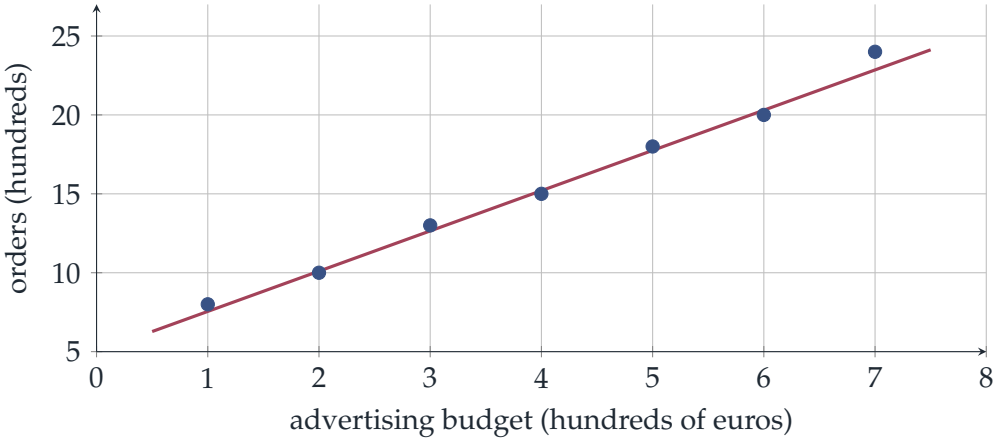
Chapter 5

Bivariate statistics

Skills: represent, choose a linear regression, interpolate, extrapolate with caution, communicate

5.1 Reference course

A bivariate data set consists of pairs (x_i, y_i). The scatter plot shows a trend, the spread and any outliers.



Key points

A line of best fit $y = ax + b$ gives estimates, not exact equalities. The coefficient a describes the estimated average change in y when x increases by one unit. Interpolation stays within the range of observed values; extrapolation goes outside that range and must be presented with caution.

The mean point $G(\bar{x}, \bar{y})$ has coordinates $\bar{x} = \dfrac{x_1 + \cdots + x_n}{n}$ and $\bar{y} = \dfrac{y_1 + \cdots + y_n}{n}$. A regression line obtained with a calculator passes through G, up to rounding.

Method

To use $y = ax + b$: 1) replace x by the requested value to estimate y; 2) solve $ax + b = y_0$ to estimate x; 3) round according to the context; 4) state whether it is interpolation or extrapolation; 5) remind that the result depends on the model.

A change of variable, for example $z = \log(y)$ or $z = \dfrac{1}{y}$, can make a scatter plot roughly linear. We then fit (x, z), and afterwards return to the variable y.

Session 31 — Scatter plot

Bivariate statistics

26 Jan 2027 • B: 2 h – C: 1 h

Session objectives

- Construct a scatter plot.
- Choose the scales.
- Identify the variables.

Fluency – 6 to 8 minutes, no calculator

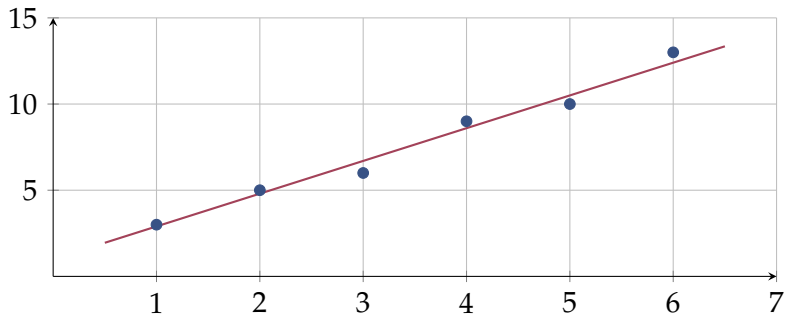
1. Write the multiplier for a 10 % increase. .....
2. Solve $3x + 4 = 25$. .....
3. Calculate the mean of 6, 8 and 10. .....
4. Give the slope between $(1, 3)$ and $(4, 9)$. .....
5. With $y = 2x + 5$, calculate y for $x = 7$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: construct a scatter plot and choose the scales.

Investigation

The table (x_i, y_i) gives the advertising spending and the sales over six months: $(1, 12), (2, 15), (3, 17), (4, 22), (5, 24), (6, 28)$. Choose scales, draw the scatter plot and plot the mean point.



Written record

A scatter plot represents the pairs (x_i, y_i). The axes, units and scales must be indicated; each pair gives a single point.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 3x + 13$, estimate y when $x = 8$.
2. Solve $3x + 13 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, course pp. 110–113; exercises p. 114, nos. 1–6. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 114, exercise 2

Plot the scatter plot of the following statistical series on a coordinate grid.

x_i	-3	0	3	5	8
y_i	1	-1	-2	5	7

Homework

Redo practice exercise 2 with the following data: replace the initial value with 275 and the rate or ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 32 — Reading and interpreting a scatter plot

Bivariate statistics

28 Jan 2027 • B: 1 h – C: 2 h

Session objectives

- Describe a trend.
- Spot spread and outliers.
- Avoid unwarranted causality.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 7 - x + 2$. .....
2. Calculate the mean of 7, 9 and 11. .....
3. Give the slope between $(1, 3)$ and $(4, 9)$. .....
4. With $y = 2x + 5$, calculate y for $x = 7$. .....
5. Solve $3x + 4 = 25$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: describe a trend and spot spread and outliers.

Investigation

Compare three scatter plots: a tight increasing trend, a scattered decreasing trend, no trend. For each, write two sentences without using the word “cause”.

Written record

Describing a scatter plot means commenting on its direction, shape, spread and outliers, without claiming a causality that has not been established.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 4x + 14$, estimate y when $x = 8$.
2. Solve $4x + 14 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, p. 116, nos. 11–15. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 114, exercise 6

The textbook graph shows the number of cigarettes sold in France as a function of the number of wolf packs recorded in France between 1995 and 2017.
How can the change in cigarette sales be described from this graph? Can we nevertheless think there is a connection between these facts?

Homework

Redo practice exercise 3 with the following data: replace the initial value with 280 and the rate or ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 33 — Linear regression

Bivariate statistics

2 Feb 2027 • B: 2 h – C: 1 h

Session objectives

- Obtain a linear regression.
- Interpret a and b.
- Check with the mean point.

Fluency – 6 to 8 minutes, no calculator

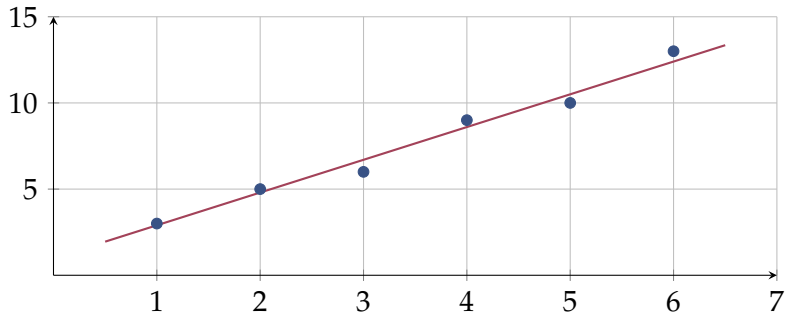
1. Calculate 20 % of 160. .....
2. Give the slope between $(1,3)$ and $(4,9)$. .....
3. With $y = 2x + 5$, calculate y for $x = 7$. .....
4. Solve $3x + 4 = 25$. .....
5. Calculate the mean of 8, 10 and 12. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: obtain a linear regression and interpret a and b.

Investigation

With the data of session 31, obtain the linear regression with a calculator. Interpret the slope and check that the mean point lies approximately on the line.



Written record

A line $y = ax + b$ fits a scatter plot. a represents the estimated average change in y for one extra unit of x.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 5x + 15$, estimate y when $x = 8$.
2. Solve $5x + 15 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, course pp. 110–113; exercises p. 117, nos. 16–18. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 115, exercise 9

1. Plot on a coordinate grid the scatter plot associated with the statistical series opposite.

x_i	-2	-1	0	2	3	5
y_i	4	8	12	16	16	20

   Draw the line with equation $y = 2x + 10$ on this grid. Does this line seem a satisfactory fit for the scatter plot? If not, suggest another one and state its reduced equation.

2. Using this equation, determine:
   - the ordinate of the point on the fitted line with abscissa 1;
   - the ordinate of the point on the fitted line with abscissa 10;
   - the abscissa of the point on the fitted line with ordinate 7.

Homework

Redo practice exercise 2 with the following data: replace the initial value with 285 and the rate or ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 34 — Interpolation

Bivariate statistics

4 Feb 2027 • B: 1 h – C: 2 h

Session objectives

- Interpolate.
- Solve using the model.
- Express an estimate.

Fluency – 6 to 8 minutes, no calculator

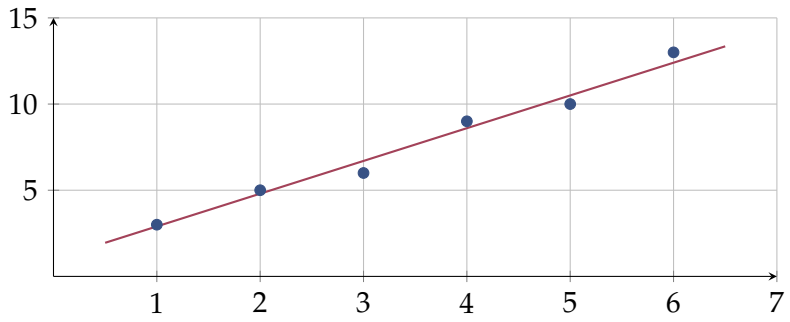
1. Write the multiplier for a 25 % increase. .....
2. With $y = 2x + 5$, calculate y for $x = 7$. .....
3. Solve $3x + 4 = 25$. .....
4. Calculate the mean of 9, 11 and 13. .....
5. Give the slope between $(1, 3)$ and $(4, 9)$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: interpolate and solve using the model.

Investigation

Over the observed period $x \in [1, 12]$, the model is $y = 2.8x + 14$. Estimate y for $x = 7.5$, then estimate x when $y = 35$. Classify each estimate.



Written record

An interpolation estimates a value inside the observed range. The result is an estimate that depends on the model.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 6x + 16$, estimate y when $x = 8$.
2. Solve $6x + 16 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, p. 118, nos. 19–24. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 115, exercise 8

The table below gives the number of new electric vehicles registered in France from 2014 to 2018.

Year	2014	2015	2016	2017	2018
Year index x_i	0	1	2	3	4
Number of thousands of vehicles y_i	10.5	17.3	21.7	24.9	31

1. Plot the scatter plot associated with this time series on a coordinate grid.
2. Suggest a linear regression for this scatter plot.
3. If we assume that this fitted model remains valid for several years, how many new vehicles would have been registered in 2020? From which year would more than 50 000 new vehicles be registered?

Homework

Redo practice exercise 3 with the following data: replace the initial value with 290 and the rate or ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 35 — Extrapolation

Bivariate statistics

9 Feb 2027 • B: 2 h – C: 1 h

Session objectives

- Extrapolate with caution.
- Assess plausibility.
- Present a limitation of the model.

Fluency – 6 to 8 minutes, no calculator

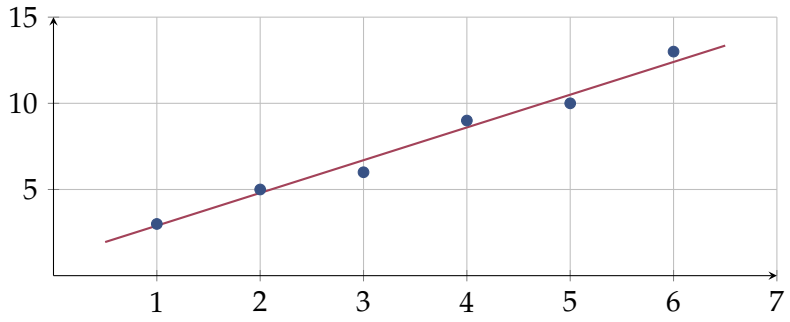
1. Simplify $3x + 3 - x + 2$. .....
2. Solve $3x + 4 = 25$. .....
3. Calculate the mean of 3, 5 and 7. .....
4. Give the slope between $(1, 3)$ and $(4, 9)$. .....
5. With $y = 2x + 5$, calculate y for $x = 7$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: extrapolate with caution and assess plausibility.

Investigation

The same model $y = 2.8x + 14$, fitted on $[1, 12]$, is used for $x = 40$. Calculate the estimate, then give three reasons why it may be unreliable.



Written record

An extrapolation estimates outside the observed range. The further away we go, the more caution is needed.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 2x + 12$, estimate y when $x = 8$.
2. Solve $2x + 12 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, p. 119, nos. 25–27. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 118, exercise 20

The annual turnover of a company (in thousands of euros) was recorded over its first five years of existence.

Year	2016	2017	2018	2019	2020
Year index x_i	0	1	2	3	4
Turnover y_i	315.6	489.9	579.3	763.5	874.5

1. Plot, on an orthogonal grid, the scatter plot $M_i(x_i; y_i)$ of this statistical series.
2. This scatter plot is fitted by the line (M_1M_5). Determine an equation of this fitted line (round the coefficients to the nearest tenth).
3. If the trend continues, what turnover can be forecast for 2024? At the end of which year will the turnover exceed two million euros?

Homework

Redo practice exercise 2 with the following data: replace the initial value with 295 and the rate or ratio with 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 36 — Change of variable

Bivariate statistics

11 Feb 2027 • B: 1 h – C: 2 h

Session objectives

- Carry out a change of variable.
- Fit the new data.
- Return to the original variable.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 5 % of 80. .....
2. Calculate the mean of 4, 6 and 8. .....
3. Give the slope between $(1,3)$ and $(4,9)$. .....
4. With $y = 2x + 5$, calculate y for $x = 7$. .....
5. Solve $3x + 4 = 25$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: carry out a change of variable and fit the new data.

Investigation

The data appear to be exponential. We set $z = \log(y)$ and the fit gives $z = 0.08x + 1.2$. Express y as a function of x, then calculate y for $x = 5$.

Written record

A change of variable can make a scatter plot linear. After fitting in the new variable, we must return to the original variable.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 3x + 13$, estimate y when $x = 8$.
2. Solve $3x + 13 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, p. 120, nos. 28–29. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 120, exercise 29

In a concert hall, sound intensity levels from a sound generator were measured at several places. The measurements were recorded in the table below.

Distance x_i (m)	0.5	1	2	3	5	10	11	12	15	20
Sound intensity y_i (dB)	97	85	73	66	59	53	55	52	55	49

A linear regression does not seem relevant, so a change of variables is considered.

1. Complete the textbook table with $u_i = \log(x_i)$.
2. Plot the scatter plot $(u_i; y_i)$ on a coordinate grid, then suggest a linear regression for it.
3. Deduce an expression for the sound intensity as a function of distance.
4. Using the expression found, determine the expected sound intensity at a distance of 8 metres, then of 30 metres.

Homework

Redo practice exercise 3 with the following data: replace the initial value with 300 and the rate or ratio with 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 37 — Choosing the model and fitting

Bivariate statistics

2 Mar 2027 • B: 2 h – C: 1 h

Session objectives

- Compare models.
- Use residuals.
- Define a range of use.

Fluency – 6 to 8 minutes, no calculator

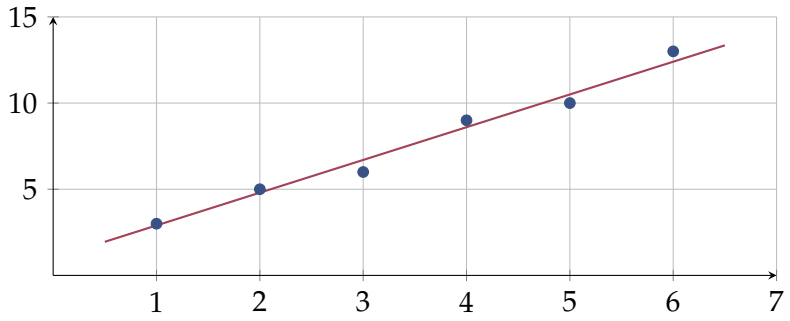
1. Write the multiplier for a 10 % increase. .....
2. Give the slope between $(1,3)$ and $(4,9)$. .....
3. With $y = 2x + 5$, calculate y for $x = 7$. .....
4. Solve $3x + 4 = 25$. .....
5. Calculate the mean of 5, 7 and 9. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: compare models and use residuals.

Investigation

Two models give total absolute residuals of 18.4 and 11.7 over the observed range. The second, however, shows unrealistic growth beyond $x = 20$. Choose a model to interpolate at $x = 12$, then discuss extrapolation at $x = 50$.



Written record

We choose a model according to the shape of the scatter plot, the size of the residuals and the range of use. A better visual fit does not allow unlimited extrapolation.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 4x + 14$, estimate y when $x = 8$.
2. Solve $4x + 14 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, p. 121, nos. 30–32, then p. 122, no. 33. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 119, exercise 25

We want to determine the caffeine concentration of a drink by measuring absorbance. To do so, a laser beam is used and the attenuation of the emitted light after passing through the liquid is measured. For the purposes of the experiment, several solutions of known caffeine content are tested and the following series of values is obtained.

Absorbance A	0.155	0.320	0.465	0.630	0.935
Concentration C (mg/L)	5	10	15	20	30

1. Represent these data with a scatter plot on a coordinate grid.
2. Does a linear regression seem relevant here? If so, express the caffeine concentration C as a function of the absorbance A.
3. An absorbance of 0.5 is measured for a drink containing caffeine. What is its caffeine concentration?

Homework

Redo practice exercise 2 with the following data: replace the initial value with 305 and the rate or ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 38 — Review and assessment

Bivariate statistics

4 Mar 2027 • B: 1 h – C: 2 h

Session objectives

- Choose and use a regression.
- Write a statistical conclusion.
- Analyse an error.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 6 - x + 2$. .....
2. With $y = 2x + 5$, calculate y for $x = 7$. .....
3. Solve $3x + 4 = 25$. .....
4. Calculate the mean of 6, 8 and 10. .....
5. Give the slope between $(1, 3)$ and $(4, 9)$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: choose and use a regression and write a statistical conclusion.

Investigation

Starting from a scatter plot supplied by the teacher, produce a complete answer: choice of model, equation, estimate, type of estimate, unit and caveat. Swap papers and use a six-criterion grid.

Written record

A complete statistical answer contains the model, the calculation, the rounding, the unit, the type of estimate and a caveat on validity.

Worked example

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Practice – essential

1. For $y = 5x + 15$, estimate y when $x = 8$.
2. Solve $5x + 15 = 50$.
3. The model was built on $[2, 12]$: classify the estimates for $x = 8$ and $x = 30$.

Textbook references

Further reading: Calao, pp. 122–123. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the

textbook.

Calao textbook – p. 118–119, exercise 24

Choose the correct statement. Justify your answer. Consider the scatter plot $M_1(1;16)$, $M_2(3;13)$ and $M_3(5;4)$.

1. The mean point of this scatter plot is:
   a. the point with coordinates $(3;10)$;
   b. the point M_2;
   c. the point with coordinates $(3;11)$.
2. The line (M_1M_3) has equation:
   a. $y = 4x + 16$;
   b. $y = -3x + 19$;
   c. $y = -2.93x + 17.7$.
3. The regression line obtained by the calculator:
   a. passes through one of the three points;
   b. is parallel to the line (M_1M_3);
   c. passes through the origin.

Homework

Redo practice exercise 3 with the following data: replace the initial value with 310 and the rate or ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.